Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

The wavelength K_α of X -rays for two metals ' A' and 'B' are $\frac{4}{1875R}$ and $\frac{1}{675R}$ respectively where ' R ' is Rydberg constant. Find the number of elements lying between A and B according to their atomic numbers

(A) 3

(B) 1

(C) 4

(D) 5

Q.2

Excitation energy of a hydrogen like ion in its excitation state is $40.8eV$. Energy needed to remove the electron from the ion in ground state is

(A) $54.4eV$

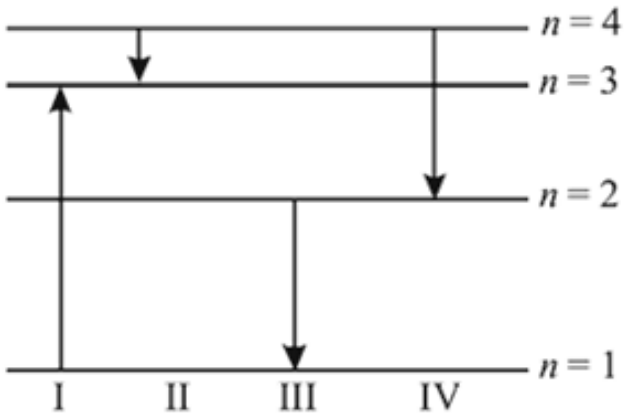
(B) $13.6eV$

(C) $40.8eV$

(D) $27.2eV$

Q.3

The diagram shows the energy levels for an electron in a certain atom. Which transition shown represents the emission of a photon with the most energy?



(A) I

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(B) II

(C) III

(D) IV

Q.4

Consider the spectral line resulting from the transition $n = 2 \rightarrow n = 1$ in the atoms and ions given below. The shortest wavelength is produced by

(A) Hydrogen atom

(B) Deuterium atom

(C) Singly ionized Helium

(D) Doubly ionised Lithium

Q.5

The momentum of hydrogen atom when a photon is emitted in a transition from $n_i = 10$ to $n_f = 1$ is

(A) $7 \times 10^{-27}\ \mathrm{kg/ms}$

(B) $7 \times 10^{27}\ \mathrm{kg/ms}$

(C) $3.5 \times 10^{-27}\ \mathrm{kg/ms}$

(D) $3.5 \times 10^{27}\ \mathrm{kg/ms}$

Q.6

A hydrogen atom makes a transition from $\mathrm{n} = 2$ to $\mathrm{n} = 1$ and emits a photon. This photon strikes a doubly ionized lithium atom $(\mathrm{z} = 3)$ in excited state and completely removes the orbiting electron. The least quantum number for the excited state of the ion for the process is:

(A) 2

(B) 4

(C) 5

(D) 3

Q.7

Suppose that a material emits X-rays of wavelengths $\lambda_{\mathrm{K}_\alpha}, \lambda_{\mathrm{K}_\beta}, \lambda_{\mathrm{L}_\alpha,}$ when it is excited by fast moving electrons; the wavelengths corresponding to $\mathrm{K}_\alpha, \mathrm{K}_\beta, \mathrm{L}_\alpha \mathrm{X}$ - rays of the material respectively. Then we can write

(A) $\lambda_{\mathrm{K}_\beta} = \lambda_{\mathrm{K}_\alpha} + \lambda_{\mathrm{L}_\alpha}$

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(B) $\sqrt{\lambda_{K_\beta}} = \sqrt{\lambda_{K_\alpha}} + \sqrt{\lambda_{L_\alpha}}$

(C) $\frac{1}{\lambda_{K_\beta}} = \frac{1}{\lambda_{K_\alpha}} + \frac{1}{\lambda_{L_\alpha}}$

(D) $\frac{1}{\sqrt{\lambda_{K_\beta}}} = \frac{1}{\sqrt{\lambda_{K_\alpha}}} + \frac{1}{\sqrt{\lambda_{L_\alpha}}}$

Q.8

Find the maximum angular speed (in $\times 10^{16} \mathrm{rad/sec}$) of the electron of a hydrogen atom in a stationary orbit.

(A) 2.1

(B) 4.1

(C) 8.5

(D) 11.6

Q.9

Assume in specific conditions only those transitions are allowed to hydrogen atoms in which the principal quantum number n changes by 2 . Find the smallest wavelength emitted by hydrogen and wavelenghth emitted by hydrogen in the visible range $(380 \mathrm{~nm}$ to $780 \mathrm{~nm})$ respectively (in nm)

(A) 103,487 nm

(B) 123,532 nm

(C) 211,410 nm

(D) 320,435 nm

Q.10

An energy of $24.6 \mathrm{eV}$ is required to remove one of the electrons from a neutral helium atom. The energy (in eV) required to remove both the electrons from a neutral helium atom is

(A) 38.2

(B) 49.2

(C) 51.8

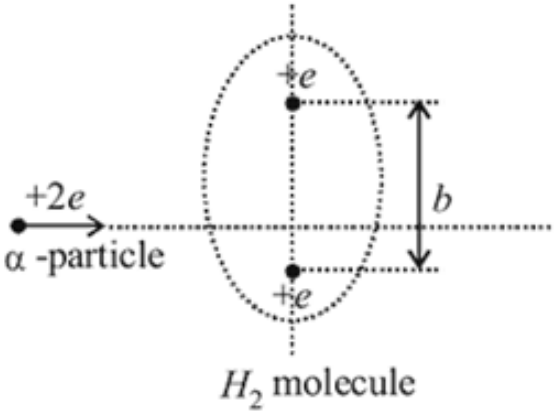
(D) 79.0

Q.11

An α particle passes rapidly through the exact centre of a hydrogen molecule, moving on a line perpendicular to the internuclear axis. The distance between the nuclei is b. Where on its path does the α

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particle experience the greatest force? (Assume that the nuclei do not move much during the passage of the α particle. Also neglect the electric field of the electrons in the molecule.



(A) $\frac{b}{2}$

(B) $\frac{b}{2\sqrt{2}}$

(C) $\frac{b}{\sqrt{2}}$

(D) None of these

Q.12

The element which has a K_αx -rays line of wavelength $1.8\mathring{A}$ is $\left(R=1.1\times10^{-7}\ \mathrm{m}^{-1}, b=1 \text{ and } \sqrt{5/33}=0.39\right)$

(A) $\mathrm{Co}, Z=27$

(B) Iron, $Z=26$

(C) $\mathrm{Mn}, Z=25$

(D) $\mathrm{Ni}, Z=28$

Q.13

Which of the following transitions in hydrogen atoms emit photons of highest frequency?

(A) $n=1$ to $n=2$

(B) $n=2$ to $n=6$

(C) $n=6$ to $n=2$

(D) $n=2$ to $n=1$

Q.14

An α -particle of $10\mathrm{MeV}$ collides head-on with a copper nucleus $(\mathrm{Z}=29)$ and is deflected back. Then, the minimum distance of approach between the centres of the two is:

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(A) 8.4×10^{-15} cm

(B) 8.4×10^{-15} m

(C) 4.2×10^{-15} m

(D) 4.2×10^{-15} cm

Q.15

Hydrogen $\left({}_1H^I\right)$, Deuterium $\left({}_1H^2\right)$, singly ionised Helium $\left({}_2He^4\right)^+$ and doubly ionised lithium $\left({}_3Li^6\right)^{++}$ all have one electron around the nucleus. Consider an electron transition from $n = 2$ to $n = 1$. If the wavelengths of emitted radiation are λ_1, λ_2, λ_3 and λ_4 respectively then approximately which one of the following is correct?

(A) $4\lambda_1 = 2\lambda_2 = 2\lambda_3 = \lambda_4$

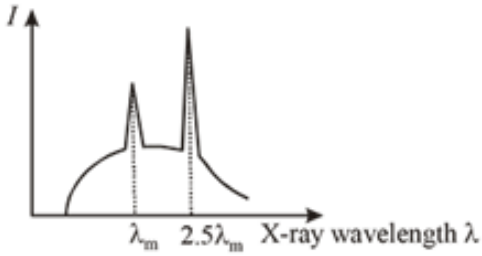
(B) $\lambda_1 = 2\lambda_2 = 2\lambda_3 = \lambda_4$

(C) $\lambda_1 = \lambda_2 = 4\lambda_3 = 9\lambda_4$

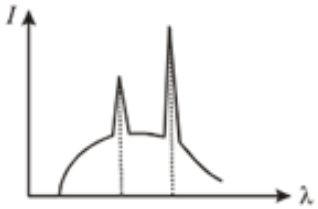
(D) $\lambda_1 = 2\lambda_2 = 3\lambda_3 = 4\lambda_4$

Q.16

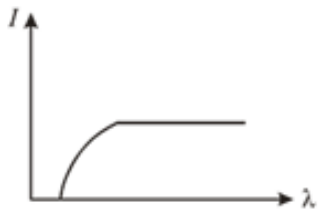
When an electron accelerated by potential difference U is bombarded on a specific metal, the emitted X-ray spectrum obtained is shown in Figure. If the potential difference is reduced to $U/3$, the correct spectrum is



(A)

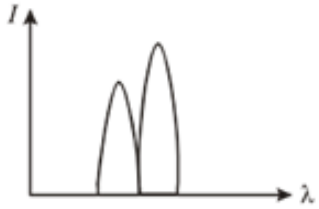


(B)



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(C)



(D)



Q.17

When the voltage applied to an X-ray tube increases from $V_1 = 10\text{kV}$ to $V_2 = 20\text{kV}$, the wavelength interval between K_α line and cut-off wavelength of continuous spectrum increase by a factor of 3. Atomic number of the metallic target is

(A) 28

(B) 29

(C) 65

(D) 66

Q.18

Ionization potential of hydrogen atom is $13.6\ \text{V}$. Hydrogen atoms in the ground state are excited by monochromatic radiation of photon energy 12.1eV. The spectral lines emitted by hydrogen atoms according to Bohr's theory will be

(A) One

(B) Two

(C) Three

(D) Four

Q.19

Which of the transition in hydrogen atom emits a photon of lowest frequency

(A) $\text{n} = 2$ to $\text{n} = 1$

(B) $n = 4$ to $n = 3$

(C) $\text{n} = 3$ to $\text{n} = 1$

(D) $n = 4$ to $n = 2$

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Q.20

Imagine an atom made up of a proton and a hypothetical particle of double the mass of the electron but having the same charge as the electron. Apply the Bohr's atom model and consider all possible transitions of this hypothetical particle to the first excited level. The longest wavelength photon that will be emitted has wavelength $\lambda(given in terms of the Rydberg constant \mathrm{R}$ for the hydrogen atom) is equal to

(A) $9/(5\mathrm{R})$

(B) $36/(5\mathrm{R})$

(C) $18/(5R)$

(D) $4/\mathrm{R}$

Answers & Solutions

Q.1 Answer:

4

Solution:

Using $\frac{1}{\lambda} = R(Z-1)^2\left[\frac{1}{n_2^2} - \frac{1}{n_1^2}\right]$

For α **particle,** $\mathrm{n}_1 = 2, \mathrm{n}_2 = 1$

For metal A:

$\frac{1875\mathrm{R}}{4} = \mathrm{R}\left(\mathrm{Z}_1 - 1\right)^2\left(\frac{3}{4}\right) \Rightarrow \mathrm{Z}_1 = 26$

For metal B:

$675\mathrm{R} = \mathrm{R}\left(\mathrm{Z}_2 - 1\right)^2\left(\frac{3}{4}\right) \Rightarrow \mathrm{Z}_2 = 31$

Therefore, 4 elements lie between A **and** B**.**

Q.2 Answer:

$54.4\mathrm{eV}$

Solution:

Excitation energy $\Delta\mathrm{E} = \mathrm{E}_2 - \mathrm{E}_1 = 13.6\mathrm{Z}^2$
$\left[\frac{1}{1^2} - \frac{1}{2^2}\right]$
$\Rightarrow 40.8 = 13.6 \times \frac{3}{4} \times \mathrm{Z}^2 \Rightarrow \mathrm{Z} = 2$

Now required energy to remove the electron from ground state

$= \frac{+13.6Z^2}{(1)^2} = 13.6(Z)^2 = 54.4\mathrm{eV}$

Q.3 Answer:

III

Solution:

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$\Delta E = E_0 \left[\frac{1}{n_1^2} - \frac{1}{n_2^2}\right]$

here (I^{st}) transition represent absorption not emission. So, (c) is correct option.

Q.4 Answer:

Doubly ionised Lithium

Solution:

No solution available

Q.5 Answer:

$7 \times 10^{-27}\ kg/ms$

Solution:

No solution available

Q.6 Answer:

4

Solution:

For this to occur, energy of transition must be more than ionization energy of lithium, hence,

$1 - \frac{1}{2^2} > 3^2 \frac{1}{n^2}$

$n > \sqrt{12}$

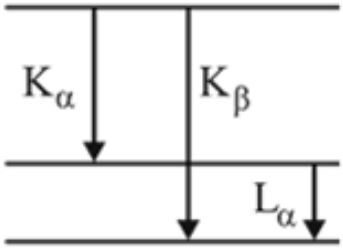
Hence, $n_{min} = 4$

Q.7 Answer:

$\frac{1}{\lambda_{K_\beta}} = \frac{1}{\lambda_{K_\alpha}} + \frac{1}{\lambda_{L_\alpha}}$

Solution:

The energy level diagram of the atom is shown in the figure. It is clear that



$E_{K_\beta} = E_{K_\alpha} + E_{L_\alpha}$

or, $v_{K_\beta} = v_{K_\alpha} + v_{L_\alpha}$

or, $\frac{1}{\lambda_{K_\beta}} = \frac{1}{\lambda_{K_\alpha}} + \frac{1}{\lambda_{L_\alpha}}$

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Q.8 Answer:

4.1

Solution:

No solution available

Q.9 Answer:

103,487 nm

Solution:

$$\mathrm{E}=13.6\left(\frac{1}{1}-\frac{1}{9}\right)=13.6\times\frac{8}{9}=\frac{\mathrm{hc}}{\lambda}$$

or $\frac{13.6\times8}{9}=\frac{4.14\times10^{-15}\times3\times10^{8}}{\lambda}$

$$\Rightarrow\lambda=\frac{4.14\times10^{-7}\times3}{13.6\times8}=103\ \mathrm{nm}$$

(b) As n changes by 2 , we may consider $\mathrm{n}=2$ **to** $\mathrm{n}=4$ **then** $\mathrm{E}=13.6\left(\frac{1}{4}-\frac{1}{16}\right)=13.6\times\frac{16-4}{64}=2.55\mathrm{eV}$ **and** $2.55=\frac{1242}{\lambda}$ **or** $\lambda=\frac{1242}{2.55}=487\ \mathrm{mm}$

Q.10 Answer:

79.0

Solution:

For helium ion, $Z=2$ **and for first orbit** $n=1$.

$E_1=\frac{-13.6}{(1)^2}\times2^2=-54.4\mathrm{eV}$

Energy required to remove this

$\mathrm{e}^{-}=+54.4\mathrm{eV}$ **Total energy required** $=54.4+24.6=79\mathrm{eV}$

Q.11 Answer:

$\frac{b}{2\sqrt{2}}$

Solution:

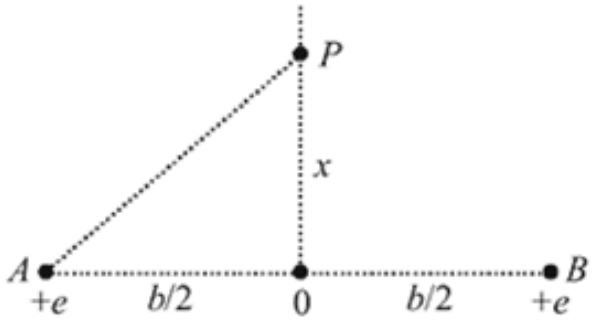
$E_p=\frac{2K_{\mathrm{ex}}}{\left(x^2+\frac{b^2}{4}\right)^{3/2}}$

For maximum E_P

$\frac{dE_P}{dx}=0$

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$\Rightarrow x = \frac{b}{2\sqrt{2}}$



Q.12 Answer:

$\mathrm{Co}, Z = 27$

Solution:

$\frac{1}{\lambda_\alpha} = \frac{3R}{4}(Z-1)^2$

$\Rightarrow \quad (Z-1) = \sqrt{\frac{4}{3R\lambda_\alpha}} = \sqrt{\frac{4}{3\times 1.1\times 10^7 \times 1.8\times 10^{-10}}}$

$= \frac{200}{3}\sqrt{\frac{5}{33}} = \frac{78}{3} = 26$

$\Rightarrow \quad Z = 26 + 1 = 27$

Q.13 Answer:

$n = 2$ to $n = 1$

Solution:

Frequency is given by

$hv = -13.6\left(\frac{1}{n_2^2} - \frac{1}{n_1^2}\right)$

For transition from $n = 6$ to $n = 2$,

$v_1 = \frac{-13.6}{h}\left(\frac{1}{6^2} - \frac{1}{2^2}\right) = \frac{2}{9} \times \left(\frac{13.6}{h}\right)$

For transition from $\mathrm{n} = 2$ to $\mathrm{n} = 1$

$v_2 = \frac{-13.6}{h}\left(\frac{1}{2^2} - \frac{1}{1^2}\right) = \frac{3}{4} \times \left(\frac{13.6}{h}\right)$

$\therefore v_1 > v_2$

Q.14 Answer:

8.4×10^{-15} m

Solution:

Given,

a. - particle have 10 Mev

at distance of closet approach, all the

KE energy of α particle converted into potential

Energy of system

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$10 \times 10^6 \text{ev} = \text{PE}$ **at distance of closest approach**

$$10 \times 10^8 \times 1.6 \times 10^{-19} \text{ J} = \frac{\text{K(ze)} \times (+2\text{e})}{\text{d}}$$
$$\text{d} = \frac{9 \times 10^9 \times 29 \times 2 \times \left(1.6 \times 10^{-19}\right)^2}{10 \times 1.6 \times 10^{-19} \times 10^6}$$
$$\text{d} = 835.2 \times 10^{-17} \text{ m}$$
$$\text{d} = 8.4 \times 10^{-15} \text{ m}$$

Q.15 Answer:

$\lambda_1 = \lambda_2 = 4\lambda_3 = 9\lambda_4$

Solution:

From Rydberg's formula, $\frac{1}{\lambda} = \text{Rz}^2 \left(\frac{1}{1^2} - \frac{1}{2^2}\right)$

$$\therefore \lambda = \frac{4}{3\text{Rz}^2}$$

$$\lambda_1 = \frac{4}{3\text{R}}$$

$$\lambda_2 = \frac{4}{3\text{R}}$$

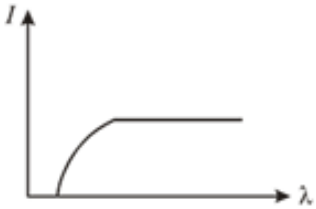
$$\lambda_s = \frac{4}{12\text{R}}$$

$$\lambda_4 = \frac{4}{27\text{R}}$$

$$\Rightarrow \lambda_1 = \lambda_2 = 4\lambda_3 = 9\lambda_4$$

Q.16 Answer:

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Solution:

λ_{m} **will increase to** $3\lambda_{\mathrm{m}}$ **due to decrease in the energy of bombarding electrons. Hence, no characteristic X-rays will be visible, only continuous X-ray will be produced.**

Q.17 Answer:

29

Solution:

The cut-off wavelength when

$\mathrm{V} = \mathrm{V}_1 = 10\mathrm{kV}$ **is**

$\lambda_1 = \frac{hc}{eV_1} = 1243.125 \times 10^{-13}\ \mathrm{m}$

The cut-off wavelength when

$\mathrm{V} = \mathrm{V}_2 = 20\mathrm{kV}$ **is**

$\lambda_2 = \frac{hc}{eV_2} = 621.56 \times 10^{-13}\ \mathrm{m}$

The wavelength corresponding to K_α **line is**

$\frac{1}{\lambda} = \frac{3R}{4}(Z-1)^2$

From given information, $(\lambda - \lambda_2) = 3\,(\lambda - \lambda_1)$

Solving above equation, we get $Z = 29$

Q.18 Answer:

Three

Solution:

Final energy of electron $= -13.6 + 12.1$

$= -1.51\mathrm{eV}$**. Which is corresponds to third level i.e.,** $\mathrm{n} = 3$**. Hence number of spectral lines emitted**

$\frac{\mathrm{n}(\mathrm{n}-1)}{2} = \frac{3(3-1)}{2} = 3$

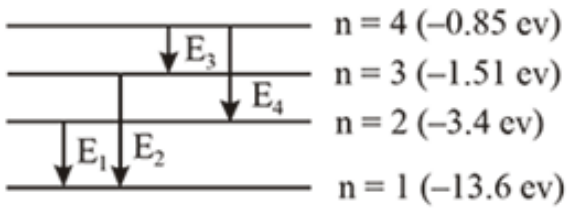
Q.19 Answer:

$n = 4$ to $n = 3$

Solution:

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From diagram



$\mathrm{E}_1 = -13.6 - (-3.4) = -10.2\mathrm{eV}$
$\mathrm{E}_2 = -13.6 - (-1.51) = -12.09\mathrm{eV}$
$\mathrm{E}_3 = -1.51 - (-0.85) = -0.66\mathrm{eV}$
$\mathrm{E}_4 = -3.4 - (-0.85) = (-2.55)\mathrm{eV}$
E_3 is least, i.e., frequency is lowest.

Q.20 Answer:

$18/(5R)$

Solution:

In hydrogen atom $\mathrm{E_n} = -\frac{\mathrm{Rhc}}{\mathrm{n}^2}$

Also $\mathrm{E_n} \propto \mathrm{m}$**; where** m **is the mass of the electron. Here the electron has been replaced by a particle**

whose mass is double of an electron. Therefore, for this hypothetical atom energy in n^{th} **orbit will be**

given by

$\mathrm{E_n} = -\frac{2\mathrm{Rhc}}{\mathrm{n}^2}$

The longest wavelength $\lambda_{\max}$ **(or minimum energy) photon will correspond to the transition of particle from** $n = 3$ **to** $n = 2$

$\frac{\mathrm{hc}}{\lambda_{\max}} = \mathrm{E_3} - \mathrm{E_2} = 2\mathrm{Rhc}\left(\frac{1}{2^2} - \frac{1}{3^2}\right)$

This gives $\lambda_{\max} = \frac{18}{5\mathrm{R}}$.

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